

HDOC Day 25 - Answer

Algorithm

1. Start and display four decreasing-pattern choices plus Exit. 2. Read the menu choice. 3. For choices 1-4, read a positive number of rows n. 4. Use nested loops only (no arrays). 5. Print the selected decreasing pattern. 6. Display an error for an invalid choice. 7. Repeat until Exit is selected.

C Program

```
#include <stdio.h>

int main() {
    int choice, n, i, j;
    do {
        printf("\n1. Decreasing right-angled triangle\n");
        printf("2. Decreasing natural-number pattern\n");
        printf("3. Decreasing repeated-number pattern\n");
        printf("4. Reverse-number triangle\n");
        printf("5. Exit\nEnter choice: ");
        scanf("%d", &choice);

        if (choice >= 1 && choice <= 4) {
            do {
                printf("Enter number of rows: ");
                scanf("%d", &n);
                if (n <= 0) printf("Rows must be positive.\n");
            } while (n <= 0);

            switch (choice) {
                case 1:
                    for (i = n; i >= 1; i--) {
                        for (j = 1; j <= i; j++) printf("* ");
                        printf("\n");
                    }
                    break;
                case 2:
                    for (i = n; i >= 1; i--) {
                        for (j = 1; j <= i; j++) printf("%d ", j);
                        printf("\n");
                    }
                    break;
                case 3:
                    for (i = n; i >= 1; i--) {
                        for (j = 1; j <= i; j++) printf("%d ", i);
                        printf("\n");
                    }
                    break;
                case 4:
                    for (i = n; i >= 1; i--) {
                        for (j = i; j >= 1; j--) printf("%d ", j);
                        printf("\n");
                    }
                    break;
                case 5: printf("Exiting program...\n"); break;
                default: printf("Invalid menu choice.\n");
            }
        } while (choice != 5);
        return 0;
    }
}
```

Expected patterns for n = 4

Choice 1:	Choice 2:	Choice 3:	Choice 4:
* * * *	1 2 3 4	4 4 4 4	4 3 2 1
* * *	1 2 3	3 3 3	3 2 1
* *	1 2	2 2	2 1
*	1	1	1

Test-case table

Test	Input	Choice	Expected result
1	n=4	1	Decreasing asterisk triangle with 4,3,2,1 stars.
2	n=4	2	Rows: 1 2 3 4; 1 2 3; 1 2; 1.
3	n=4	3	Rows: 4 4 4 4; 3 3 3; 2 2; 1.
4	n=4	4	Rows: 4 3 2 1; 3 2 1; 2 1; 1.
5	n=-2	1	Rows must be positive; asks again.